



UNIT–III: Cloud Deployment Models

TYBCA • Textbook-Aligned & University-Exam Oriented

Reference: Thomas Erl, Ricardo Puttini & Zaigham Mahmood

Learning Objectives

Explain cloud deployment models.

Describe Public, Private, Hybrid and Community Clouds.

Identify ownership, access and management characteristics.

Compare the four deployment models.

Choose an appropriate model for a given situation.

Prepare structured 5-mark and 8-mark answers.

1. Cloud Deployment Model

Describes how a cloud environment is organized and made available to consumers.

Concerned with ownership, access and deployment of the cloud environment.

Do not confuse deployment models with service models.

Service model asks: What is provided? → IaaS, PaaS, SaaS.

Deployment model asks: Who owns/uses/accesses the cloud environment?

Service Model vs Deployment Model

Service model: what cloud capability/service is provided.

Deployment model: how the cloud environment is organized and who can access it.

Service models: IaaS, PaaS, SaaS.

Deployment models: Public, Private, Hybrid, Community.

Exam tip: clearly distinguish these two concepts.

2. Four Base Cloud Deployment Models

Public Cloud

Private Cloud

Hybrid Cloud

Community Cloud

Memory aid: Public = Provider; Private = Organization; Community = Group; Hybrid = Combination.

3. Public Cloud — Definition

A public cloud is a publicly accessible cloud environment owned by a third-party cloud provider.

The provider creates and maintains the public cloud and its IT resources.

IT resources are generally provisioned using cloud delivery models.

Services are commonly offered commercially.

Public Cloud — Key Characteristics

Ownership: third-party cloud provider.

Access: publicly accessible to cloud consumers.

Provider responsibility: creation and ongoing maintenance.

Resources: provisioned as cloud IT resources.

Commercialization: generally offered at a cost or through other commercial avenues.

Public Cloud — Exam Answer

Definition and ownership.

Accessibility and provider responsibility.

How resources are made available.

Commercial nature.

Draw provider → multiple consumers diagram.

Conclude: provider-owned cloud made available to external consumers.

4. Private Cloud — Definition

A private cloud is owned by a single organization.

It enables the organization to centralize access to IT resources.

Different departments, locations or parts of the organization can access resources.

The environment can be managed internally or through outsourced staff.

Private Cloud — Organization Roles

The same organization may technically perform both provider and consumer roles.

A cloud-provisioning department can provide cloud resources.

Other departments can consume those resources.

This creates centralized access to IT resources across the organization.

Private Cloud — Important Concept

A private cloud may physically reside on an organization's premises.

Resources can still be considered cloud-based when made remotely accessible to cloud consumers.

Therefore, on-premise location alone does not determine whether an environment is cloud.

Exam tip: focus on how resources are provided and consumed.

Private Cloud — Exam Answer

Definition

Single-organization ownership

Centralized access to IT resources

Provider and consumer roles

Management arrangement

Suitable diagram and example/situation

5. Hybrid Cloud — Definition

A hybrid cloud is a cloud environment composed of two or more different cloud deployment models.

A common combination is private cloud + public cloud.

Different workloads can be placed in different environments.

Example: sensitive services in private cloud and less-sensitive services in public cloud.

Hybrid Cloud — Working

Identify workloads with different requirements.

Place sensitive services/workloads in a private environment.

Place less-sensitive services/workloads in a public environment.

Use the environments as a combined deployment.

Result: a hybrid cloud deployment.

Hybrid Cloud — Challenges

Hybrid deployment architectures can be complex.

They can be challenging to create and maintain.

Different cloud environments may have disparities.

Management responsibilities may be split between organizations/providers.

Exam point: hybrid provides flexibility but can increase management complexity.

6. Community Cloud — Definition

A community cloud is similar to a public cloud but limits access to a specific community of cloud consumers.

The community may have common requirements.

A community cloud may be jointly owned by community members.

It may also be owned by a third-party cloud provider with restricted access.

Community Cloud — Key Characteristics

Access is limited to a specific community.

Community members may share responsibility for defining and evolving the cloud.

Outside parties are generally not granted access unless permitted.

Ownership may be shared or handled by a third-party provider.

Key distinction: community access is restricted, unlike general public access.

Community Cloud — Exam Answer

Definition

Specific community access

Possible ownership arrangements

Shared responsibility

Restriction on outside access

Suitable diagram and example/situation

7. Comparison of Deployment Models

Public: third-party provider; publicly accessible.

Private: single organization; organization-specific access.

Community: specific community; restricted access.

Hybrid: two or more deployment models combined.

Use ownership, access, management and main purpose as comparison headings.

Public vs Private Cloud

Public: owned by a third-party provider.

Private: owned by a single organization.

Public: made available to external cloud consumers.

Private: resources are centralized for the organization.

Key difference: ownership.

Private vs Community Cloud

Private: dedicated to one organization.

Community: access is limited to a specific community.

Private ownership is organization-specific.

Community ownership may be shared or handled by a third-party provider.

Key difference: intended consumer group.

Public vs Community Cloud

Public: publicly accessible cloud environment.

Community: restricted to a specific community.

Public: third-party provider commonly owns and maintains it.

Community: may involve shared responsibility among members.

Key difference: broad access vs community-restricted access.

Hybrid vs Other Deployment Models

Hybrid combines two or more deployment models.

Example: private + public cloud.

Different workloads can be assigned to different environments.

Hybrid deployments can increase architectural and management complexity.

Key idea: combination rather than a single deployment type.

8. Choosing the Right Deployment Model

Consider who needs access.

Consider ownership requirements.

Consider whether the cloud is for one organization or a specific community.

Consider workload/data sensitivity.

Consider management responsibilities and operational complexity.

Consider whether different workloads require different environments.

Choosing the Model — Quick Decision

Third-party provider + broad external availability → Public Cloud.

Single organization → Private Cloud.

Specific community with common requirements → Community Cloud.

Two or more deployment models required → Hybrid Cloud.

Sensitive + less-sensitive workloads needing different environments → Hybrid Cloud.

9. Important Exam Definitions

Public Cloud: publicly accessible cloud environment owned by a third-party provider.

Private Cloud: cloud owned by a single organization for centralized access to IT resources.

Community Cloud: cloud with access limited to a specific community of cloud consumers.

Hybrid Cloud: cloud environment composed of two or more different deployment models.

10. Expected 2-Mark Questions

Define cloud deployment model.

What is a public cloud?

What is a private cloud?

Define community cloud.

Define hybrid cloud.

Who owns a public cloud?

What is meant by a specific community?

What is a hybrid deployment model?

11. Expected 5-Mark Questions

Explain Public Cloud with its characteristics.

Explain Private Cloud with its characteristics.

Explain Community Cloud.

Explain Hybrid Cloud with a suitable example.

Explain factors to consider while choosing a deployment model.

Differentiate Public Cloud and Private Cloud.

12. Expected 8-Mark Questions

Explain the different cloud deployment models with suitable diagrams.

Explain Public Cloud and Private Cloud and differentiate them.

Explain Community Cloud and Hybrid Cloud with examples.

Compare Public, Private, Community and Hybrid Cloud.

Explain how to choose an appropriate cloud deployment model.

8-Mark Answer-Writing Formula

Definition → Ownership → Access → Characteristics → Structure/Working → Example → Challenges/Advantages → Diagram → Conclusion.

For comparison: Introduction → comparison table → key differences → suitable situations → conclusion.

Use headings and bullet points instead of one long paragraph.

Draw a simple labelled diagram wherever appropriate.

Quick Revision — Four Words

PUBLIC = PROVIDER

PRIVATE = ONE ORGANIZATION

COMMUNITY = SPECIFIC GROUP

HYBRID = COMBINATION

Use these four words to recall the core distinction.

Unit-III Final Revision

Deployment model describes how the cloud environment is organized and made available.

Public Cloud → third-party provider + public accessibility.

Private Cloud → single organization.

Community Cloud → specific community.

Hybrid Cloud → two or more deployment models.

Selection depends on access, ownership, sensitivity, management and complexity.